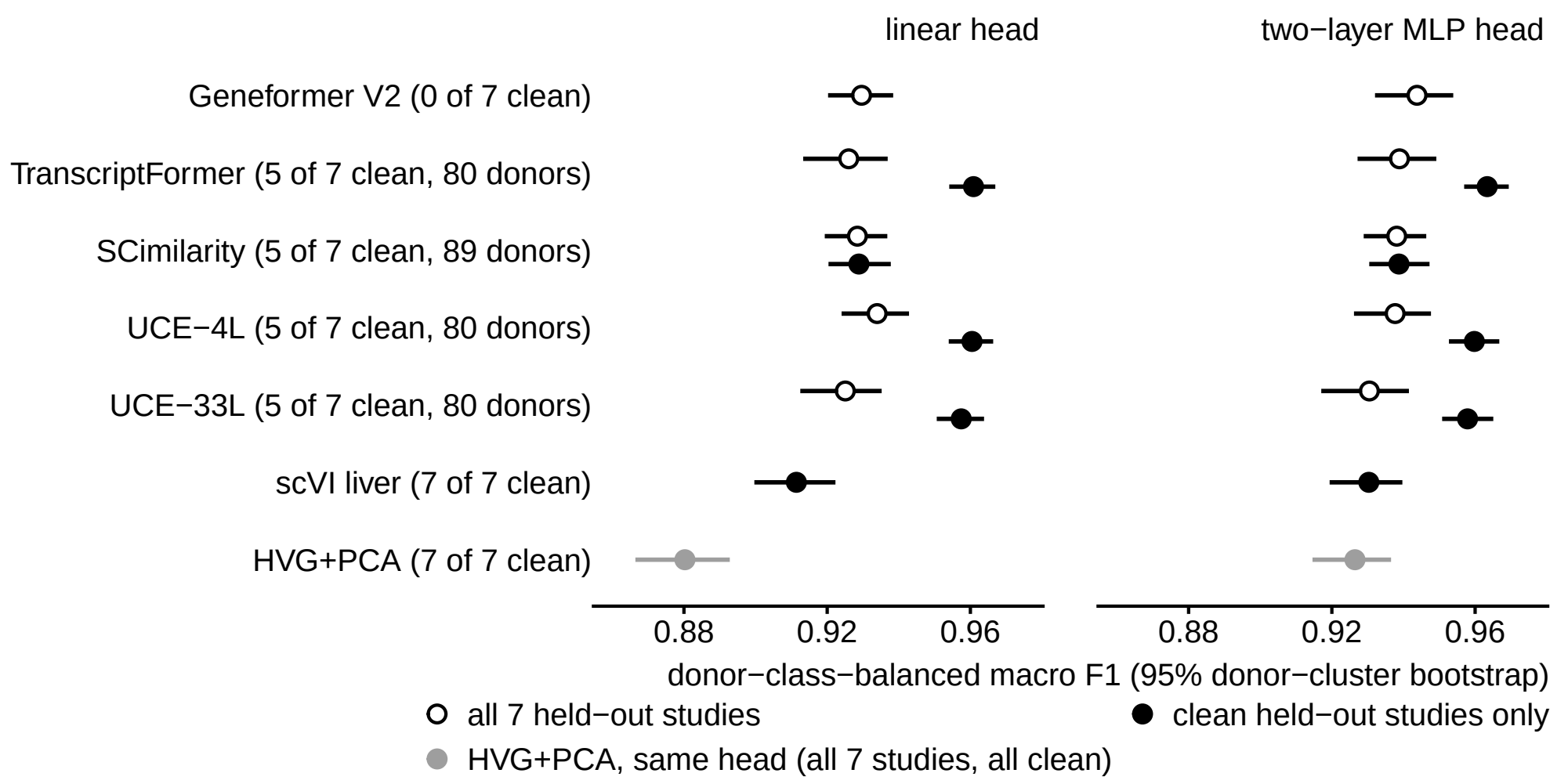
linear head
two−layer MLP head
Geneformer V2 (0 of 7 clean)
TranscriptFormer (5 of 7 clean, 80 donors)
SCimilarity (5 of 7 clean, 89 donors)
UCE−4L (5 of 7 clean, 80 donors)
UCE−33L (5 of 7 clean, 80 donors)
scVI liver (7 of 7 clean)
HVG+PCA (7 of 7 clean)
0.88
0.92
0.96
0.88
0.92
0.96
donor−class−balanced macro F1 (95% donor−cluster bootstrap)
all 7 held−out studies
clean held−out studies only
HVG+PCA, same head (all 7 studies, all clean)